

NovaTrust Bank

Marketing Campaign Performance & Budget Analysis

Internal Business Case: Campaign Spend Analysis

Report Details

- **Prepared For:** Abdelrahman Mohamed (Head of Marketing Operations)
- **Prepared By:** Data Analysis Team
- **Analysis Scope:** Review of 45,211 client contacts with an estimated total spend of €270,000.

Executive Summary

Following the Marketing Operations team's request to evaluate the recent direct phone campaign cycles (totaling €270,000 in spend), we have developed an interactive descriptive analytics dashboard. The analysis of 45,211 client contacts reveals an overall conversion rate of approximately 11.7%. However, Cost Per Acquisition (CPA) varies drastically across different demographic segments and contact strategies. This report outlines where the bank achieved actual value for money, highlights areas of budget wastage, and provides data-driven, actionable recommendations for the upcoming campaign cycle.

1 Phase 1: Data Cleaning and Transformation

To ensure the highest accuracy of the dashboard metrics, the raw data underwent rigorous transformation using Power Query:

- **Handling Missing Values:** Entries labeled as "unknown" in categorical columns (e.g., *job*, *marital*, *education*, *loan*) were replaced with "Not Specified" to maintain a professional appearance in visuals without losing valuable row data.
- **Resolving the 'pdays' Anomaly:** The value 999 in the *pdays* column (indicating a client was not previously contacted) heavily skewed numerical averages. A new conditional column, *Contact History*, was created to classify clients as either "New Contact" or "Previous Contact".
- **Target Variable Optimization:** The campaign outcome column (*y*) was converted into a binary numeric column named *Subscription_Flag* (1 for "yes", 0 for "no") to streamline the calculation of conversion rates in DAX.

2 Phase 2: Core DAX Measures

The following Key Performance Indicators (KPIs) were engineered to drive the interactive dashboard calculations:

Primary Calculations

- **Total Calls** = `COUNTROWS('CampaignData')`
- **Total Subscriptions** = `SUM('CampaignData'[Subscription_Flag])`
- **Conversion Rate %** = `DIVIDE([Total Subscriptions], [Total Calls], 0)`
- **Total Cost (€)** = `[Total Calls] * 6`
- **CPA (€)** = `DIVIDE([Total Cost], [Total Subscriptions], 0)`

3 Phase 3: Dashboard Architecture

The Power BI dashboard was structured into two primary interactive views tailored for a non-technical leadership audience:

3.1 View 1: Campaign Performance Overview

Designed to answer "How did the campaign perform across time, segments, and channels?"

- **KPI Cards:** Displaying Total Calls, Total Subscriptions, and Conversion Rate.
- **Trend Line Chart:** Month-over-month performance, highlighting the relationship between call volume and actual conversion rates.
- **Donut Chart:** Comparing the effectiveness of *Cellular* versus *Telephone* contacts.
- **Bar Chart:** Conversion rates segmented by client occupation (*job*), sorted in descending order to identify the most responsive demographics.

3.2 View 2: Budget Analysis & Efficiency

Designed to answer "Where did we get value for money, and where did we waste it?"

- **Budget Wastage Scatter Plot:** Correlating Total Cost (X-axis) with Conversion Rate (Y-axis) by occupation. Bubble size represents Total Calls. This visual instantly identifies segments that drain the budget with zero return (bottom-right quadrant).
- **Historical CPA Bar Chart:** Evaluating the Cost Per Acquisition based on the outcome of previous campaigns (*Success, Failure, Nonexistent*).
- **Detailed Matrix Table:** A drill-down table showing CPA across all segments with conditional formatting (Red for high CPA, Green for low CPA).

4 Phase 4: Key Insights

The analysis uncovered several critical patterns that management needs to consider:

1. Budget Wastage (High CPA Segments)

A massive portion of the €270,000 budget was exhausted by contacting clients in *admin* and *blue-collar* roles, particularly during the month of *May*. Despite the extremely high call volume (and consequently high cost), the conversion rate for these segments was negligible, leading to an unsustainably high Cost Per Acquisition (CPA).

2. Value for Money (Low CPA Segments)

Conversely, *Students* and *Retired* clients yielded the highest subscription rates with a fraction of the total calls, representing the best return on investment. Furthermore, months like *March* and *September* saw lower overall call volumes but vastly superior conversion efficiency.

3. Channel and History Impact

- **Cellular Supremacy:** Reaching out via mobile networks (Cellular) drastically outperforms traditional landline (Telephone) contacts.
- **The "Success" Multiplier:** Clients who successfully subscribed in a previous campaign (*poutcome* = *success*) have an incredibly low CPA. They require far less effort and cost to convert again.

5 Phase 5: Actionable Recommendations

Based on the data, we strongly recommend the following strategic pivots before approving the budget for the next campaign cycle:

1. **Eliminate Dead-End Contacts:** Immediately remove clients whose previous campaign outcome was "Failure" from the new call lists. Continuing to call them burns agent time and inflates CPA with almost zero probability of conversion.
2. **Demographic Budget Reallocation:** Defund the outreach to *blue-collar* and *services* segments. Redirect those funds to aggressively target the *Student* and *Retired* populations, where the propensity to open a term deposit is highest.
3. **Mandate Cellular-First Outreach:** Allocate at least 90% of the telecommunication budget to cellular contacts. Landline calls should be phased out as they no longer yield a justifiable ROI.
4. **Optimize Campaign Scheduling:** Avoid front-loading the budget into the month of *May*, which historically shows high activity but poor results. Instead, distribute the budget strategically across Q1 and Q3 (focusing on *March* and *September*), aligning with peak client responsiveness.

End of Report

The Analytics Team is available to present these findings and demonstrate the interactive dashboard to the Retail Banking Director upon request.